

Lecture Notes: Interest Rates (Chapter 6)

Corporate Finance

I. Introduction to Debt and Interest Rates

Investment in **debt** has a contractual maturity, unlike equity securities. The return on debt comes in the form of **interest rates**.

Different interest rates exist in the economy for different instruments:

- Bond market rates
- Money market rates (T-bills)
- Auto loans
- Commercial mortgages

This chapter examines how these interest rates are determined and how they function.

II. Four Key Factors Affecting Interest Rates

While the overall interest rate in an economy is determined by the **demand and supply of funds**, four specific factors affect interest rates for debt and business:

1. Productive Investment Opportunities

- Companies borrow to invest in projects that generate cash flows.
- The return generated on an investment acts as a **ceiling or cap** on the interest rate a company can pay.
- If a project generates a 12% return, the company cannot pay an interest rate higher than 12% and still maintain a profit margin.

2. Time Preference for Consumption

- This relates to the willingness to defer consumption (save) versus consume today.
- **Lower savings** = higher consumption today = less funds available for investment.
- To induce someone to defer consumption, they must be compensated with a **higher interest rate**.

Example: Time Preference

30-Year-Old (Newly Married with Children)	50-Year-Old (Established)
High expenses, low savings capacity	House paid off, car purchased, retirement savings
Requires higher interest rate to defer consumption	Can defer consumption at a lower interest rate

3. Risk

- This is the most significant factor in finance.
- Higher inherent risk requires compensation in the form of a **higher required return**.
- **Risk Hierarchy:**
 - T-bills: Lowest rates (no default risk)
 - Corporate bonds: Higher rates (default risk)
 - Junk bonds: Highest yields (high default risk)

4. Inflation

- Inflation is the rate of change of prices (CPI).
- Investors demand compensation to maintain purchasing power.
- If expected inflation is 10%, investors demand at least a 10% return.
- Rising inflation → rising interest rates → increased government borrowing → further upward pressure on rates.

III. Market Dynamics: Demand and Supply of Funds

Interest rates are determined by the **demand and supply of loanable funds**.

Supply and Demand Curves

- **Supply Curve:** Upward sloping. Higher interest rates incentivize more savings and supply of funds.
- **Demand Curve:** Downward sloping. Lower interest rates make borrowing cheaper, increasing demand for funds.
- **Equilibrium:** The intersection of demand and supply determines the equilibrium interest rate.

Flight to Quality

During periods of uncertainty (financial crisis, war), investors shift preferences toward low-risk securities.

Flight to Quality Dynamics

Low-Risk Market (e.g., T-bills)	High-Risk Market (e.g., Junk Bonds)
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Supply of funds ↑

Supply of funds ↓

Interest rate ↓ (e.g., 5% → 4%)

Interest rate ↑ (e.g., 7% → 8%)

Result: Risk premium widens from 2% to 4% due to increased uncertainty.

IV. Nominal vs. Real Interest Rates

- **Nominal Rate:** The quoted rate (e.g., 12% from a bank).
- **Real Rate:** The nominal rate adjusted for inflation.

$$\text{Real Return} \approx \text{Nominal Rate} - \text{Inflation Rate}$$

Key Insight

If inflation erodes purchasing power, real income may decrease even with positive nominal returns.

Banking Context in Pakistan

- Banks are earning multi-billion dollar profits due to high **spreads** (difference between lending and deposit rates).
- Islamic banks (e.g., Meezan Bank) offer 0% interest on current accounts (framed as Riba-free), then invest those funds in government T-bills earning high returns.
- This lowers their cost of funds significantly.
- Due to extraordinary profits, the government has imposed a **Super Tax** on banks.

V. Determinants of Interest Rates: The Formula

The nominal interest rate on any security is built from several components:

$$r = r^* + IP + DRP + LP + MRP$$

	Component	Description
Where:	r^*	Real Risk-Free Rate: Equilibrium rate without inflation
	IP	Inflation Premium: Compensates for expected inflation
	DRP	Default Risk Premium: Risk of non-payment (zero for Treasuries)
	LP	Liquidity Premium: Compensation for illiquidity
	MRP	Maturity Risk Premium: Compensation for price sensitivity to rate changes

Note on Treasury Securities

Treasury securities have **no default risk** ($DRP = 0$) because the government can tax or print money. However, they may default on foreign currency obligations.

VI. Practice Problem

Problem Statement

An analyst has the following information:

- Real Rate (r^*): 2% (constant for 3 years)
- Expected Inflation: Year 1 = 3%; Year 2 = 3.5%; Year 3 = 4%
- MRP Formula: $0.1 \times (t - 1)\%$ (where t = years to maturity)
- 3-Year Corporate Bond: $LP = 0.2\%$; $DRP = 0.6\%$

Calculate:

1. Rate for a 1-year T-bill
2. Rate for a 3-year T-bond
3. Rate for a 3-year Corporate bond

Solution

Step 1: Calculate Inflation Premiums (IP)

The inflation premium for a t -year security is the average of expected inflation over the t years.

$$IP_1 = 3.00\%$$

$$IP_3 = \frac{3\% + 3.5\% + 4\%}{3} = \frac{10.5\%}{3} = 3.50\%$$

Step 2: Calculate Maturity Risk Premiums (MRP)

$$MRP = 0.1 \times (t - 1)\%$$

$$MRP_1 = 0.1 \times (1 - 1)\% = 0.1 \times 0\% = 0.00\%$$

$$MRP_3 = 0.1 \times (3 - 1)\% = 0.1 \times 2\% = 0.20\%$$

Step 3: Calculate 1-Year T-bill Rate

For T-bills: $DRP = 0$, $LP = 0$

$$\begin{aligned} r_{T-bill} &= r^* + IP_1 + MRP_1 + DRP + LP \\ &= 2\% + 3.00\% + 0.00\% + 0\% + 0\% \\ &= \mathbf{5.00\%} \end{aligned}$$

Step 4: Calculate 3-Year T-bond Rate

For T-bonds: $DRP = 0$, $LP = 0$

$$\begin{aligned} r_{T-bond} &= r^* + IP_3 + MRP_3 + DRP + LP \\ &= 2\% + 3.50\% + 0.20\% + 0\% + 0\% \\ &= \mathbf{5.70\%} \end{aligned}$$

Step 5: Calculate 3-Year Corporate Bond Rate

For corporate bonds: $DRP = 0.6\%$, $LP = 0.2\%$

$$\begin{aligned} r_{Corp} &= r^* + IP_3 + MRP_3 + DRP + LP \\ &= 2\% + 3.50\% + 0.20\% + 0.6\% + 0.2\% \\ &= \mathbf{6.50\%} \end{aligned}$$

Final Answer

Results	
Security	Interest Rate
1-Year T-bill	5.00%
3-Year T-bond	5.70%
3-Year Corporate Bond	6.50%

Summary: Key Relationships

- Interest rates are determined by demand and supply of loanable funds.
- Four key factors: Investment opportunities, time preference, risk, and inflation.
- Flight to quality widens risk premiums during uncertainty.
- Nominal rate = Real rate + Inflation + Default risk + Liquidity + Maturity risk.